

EXPERIMENT 23. Draw a Line Using Axis Values Taken from a Text File

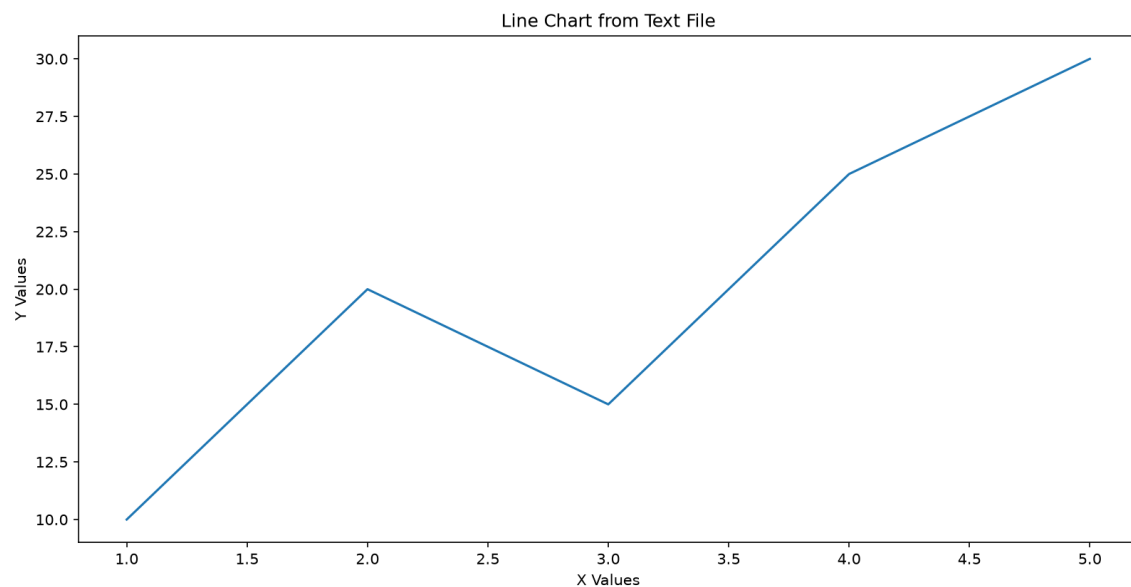
AIM

To write a Python program to read axis values from a text file and draw a line chart with suitable labels and title.

CODE

```
exp23.py > ...
1  import matplotlib.pyplot as plt
2  data = """1 10
3  2 20
4  3 15
5  4 25
6  5 30"""
7  with open("data.txt", "w") as file:
8      file.write(data)
9  x = []
10 y = []
11 with open("data.txt", "r") as file:
12     for line in file:
13         a, b = map(float, line.split())
14         x.append(a)
15         y.append(b)
16 plt.plot(x, y)
17 plt.xlabel("X Values")
18 plt.ylabel("Y Values")
19 plt.title("Line Chart from Text File")
20 plt.show()
```

OUTPUT



RESULT

Thus, the line chart was successfully created using axis values obtained from a text file.

EXPERIMENT 24. Draw Line Charts of Financial Data

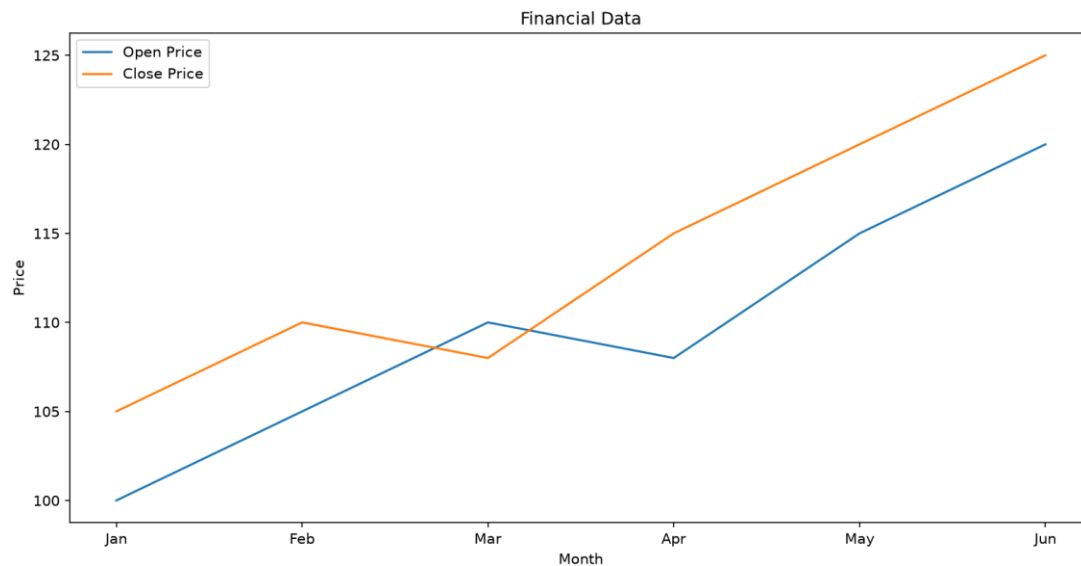
AIM

To write a Python program to draw line charts representing financial data.

CODE

```
exp24.py > ...
1  import matplotlib.pyplot as plt
2  date = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']
3  open_price = [100, 105, 110, 108, 115, 120]
4  close_price = [105, 110, 108, 115, 120, 125]
5  plt.plot(date, open_price, label='Open Price')
6  plt.plot(date, close_price, label='Close Price')
7  plt.xlabel("Month")
8  plt.ylabel("Price")
9  plt.title("Financial Data")
10 plt.legend()
11 plt.show()
12
```

OUTPUT



RESULT

Thus, the financial data was successfully represented using line charts.